What happens if you whisper a rumor to 100 AI investors?

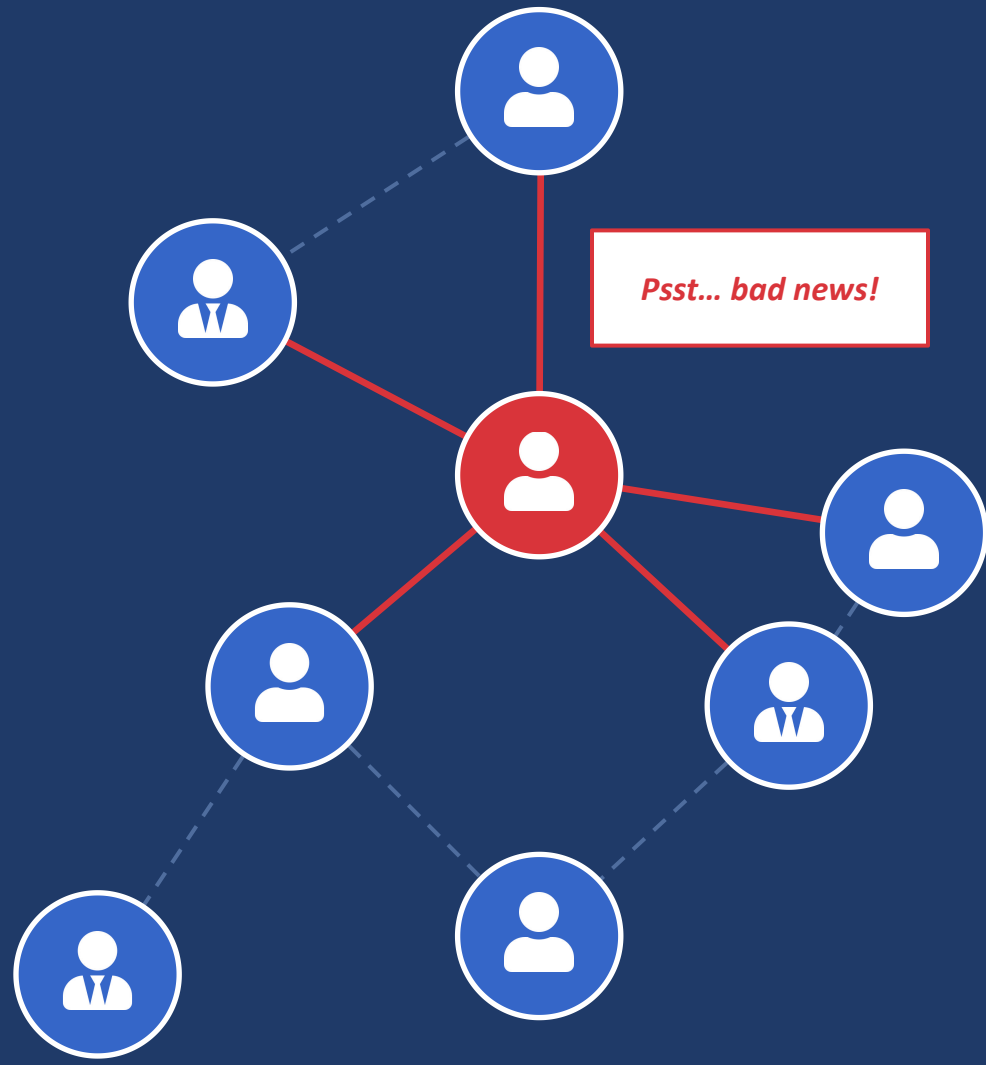
TwinMarket: A Scalable Behavioral and Social Simulation for Financial Markets

Yang, Zhang, Wu et al. · CUHK-Shenzhen & Nanjing University

NeurIPS 2025

Best Paper · ICLR 2025 Financial AI Workshop

李婧雅 Jingya Li



Markets are moved by people



Different investors, different minds

Traditional models

- Identical agents
- Fixed, simple rules
- No social media

vs

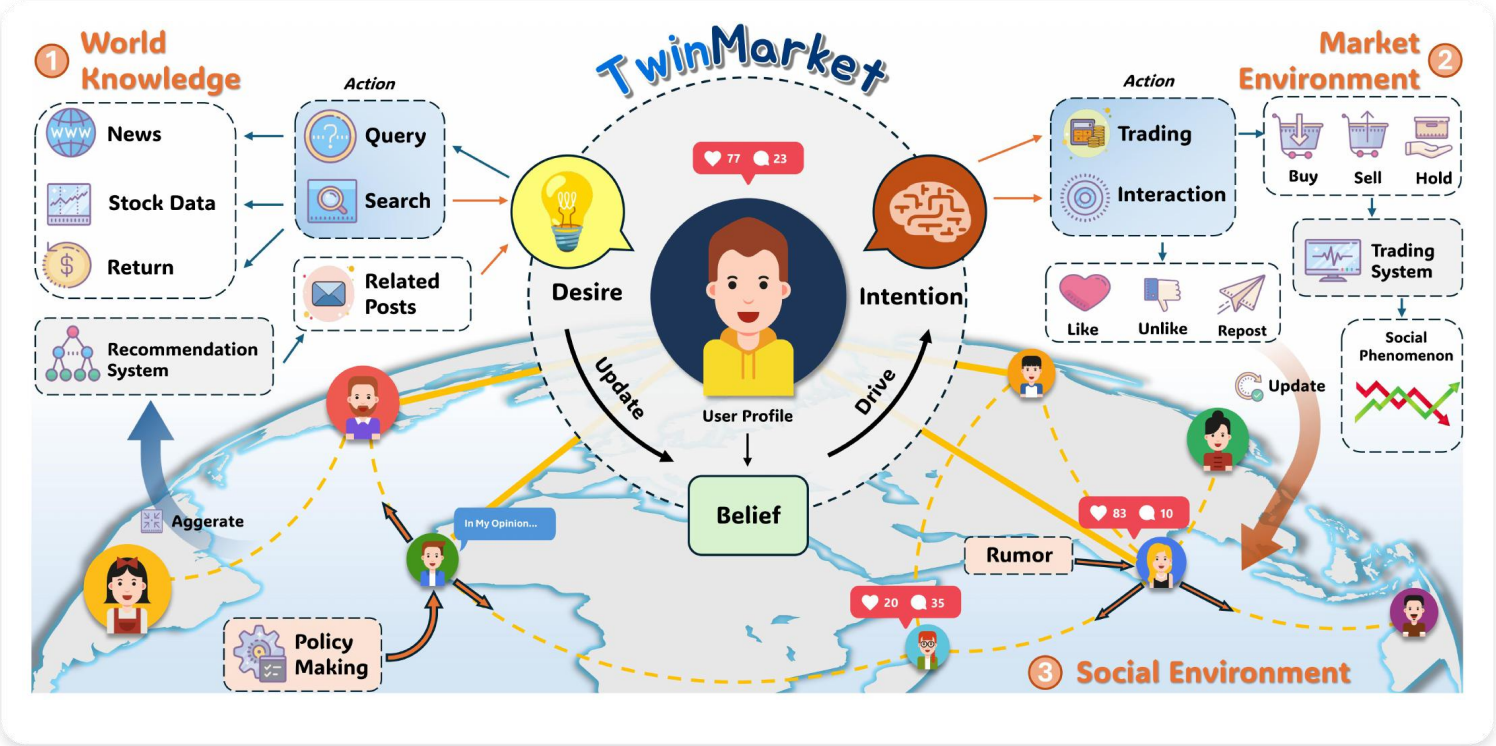
Real investors

- Emotions & biases
- Follow the crowd
- Read each other's posts

Research question: Can AI agents act like real investors, and recreate how real markets behave?

Figures & data: Yang et al., TwinMarket (NeurIPS 2025)

TwinMarket: AI investors living on a social network



Daily loop: read news & posts → search for more → update belief → trade → post → market updates

Real people as templates
639 real investors from Xueqiu set each agent's style and biases

Believe → Want → Act
Each day: form a view, search news & posts, then trade

A living social network
Similar traders see each other's posts; daily auction matches trades

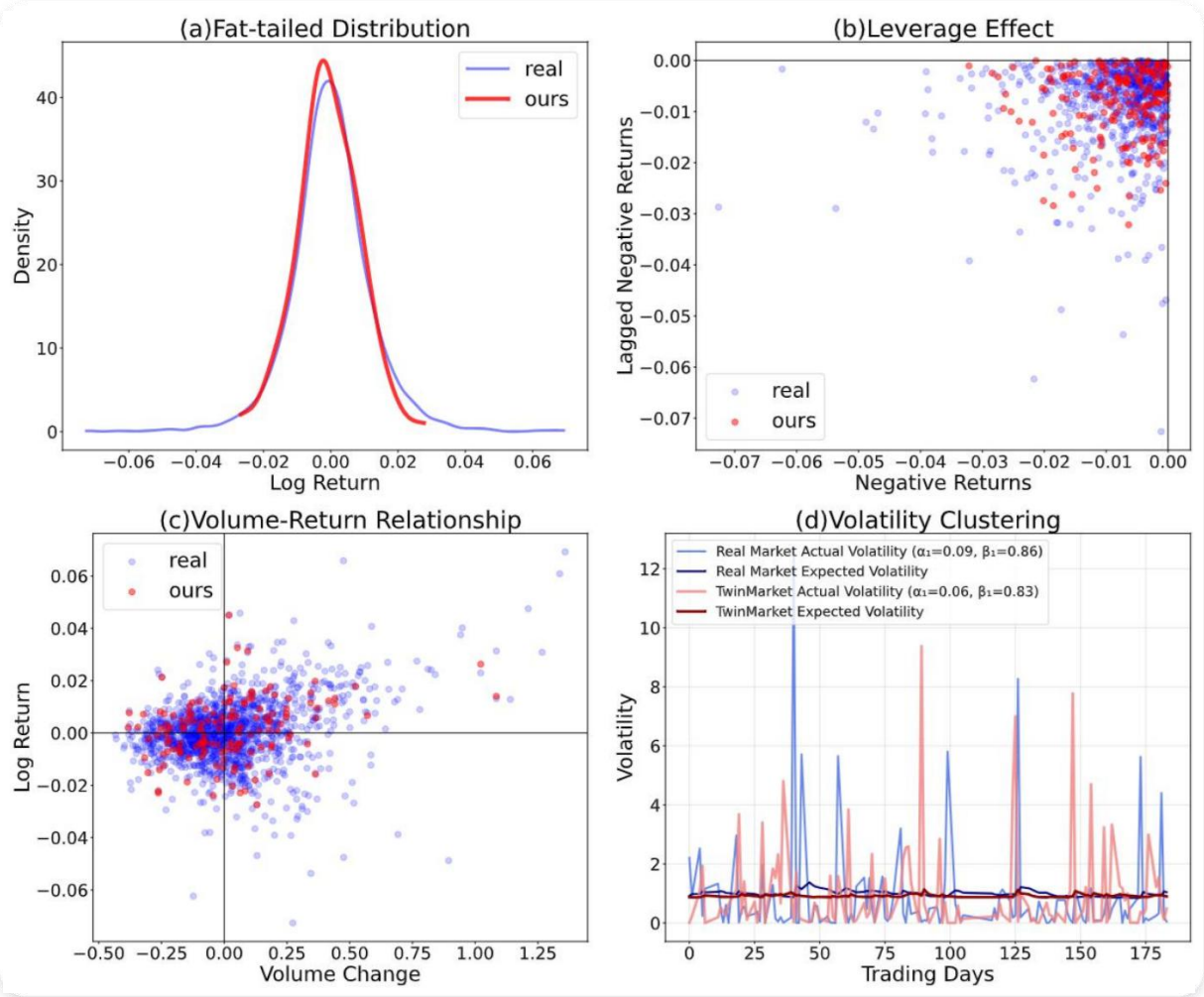
100 GPT-4o agents

5 months simulated trading

1,000 agents when scaled up

Figures & data: Yang et al., TwinMarket (NeurIPS 2025)

The simulated market behaves like the real one



Red = TwinMarket **Blue** = real market

All 4 classic market patterns reproduced

Market pattern	Real	TwinMarket
Extreme moves (kurtosis)	7.26	5.24
Volatility clustering	0.95	0.89
Leverage effect	0.14	0.11

AI investors overtrade and lose, just like humans

+6.65%

Top 10%
trade less
(turnover 4.02%)

−10.52%

Bottom 50%
trade more
(turnover 7.03%)

Figures & data: Yang et al., TwinMarket (NeurIPS 2025)

Bubbles form and burst on their own



***Blue** = average AI investor belief **Red** = price index*

1. **Optimism feeds itself**
   Good mood → buying → higher prices → even better mood

2. **Belief peaks first**
   Confidence turns around Jul 5, while prices keep climbing

3. **The crash**
   Prices follow belief down from Jul 10, wiping out the gains

Nobody programmed a bubble. It emerged from the agents.

Figures & data: Yang et al., TwinMarket (NeurIPS 2025)

One rumor, and the market panics

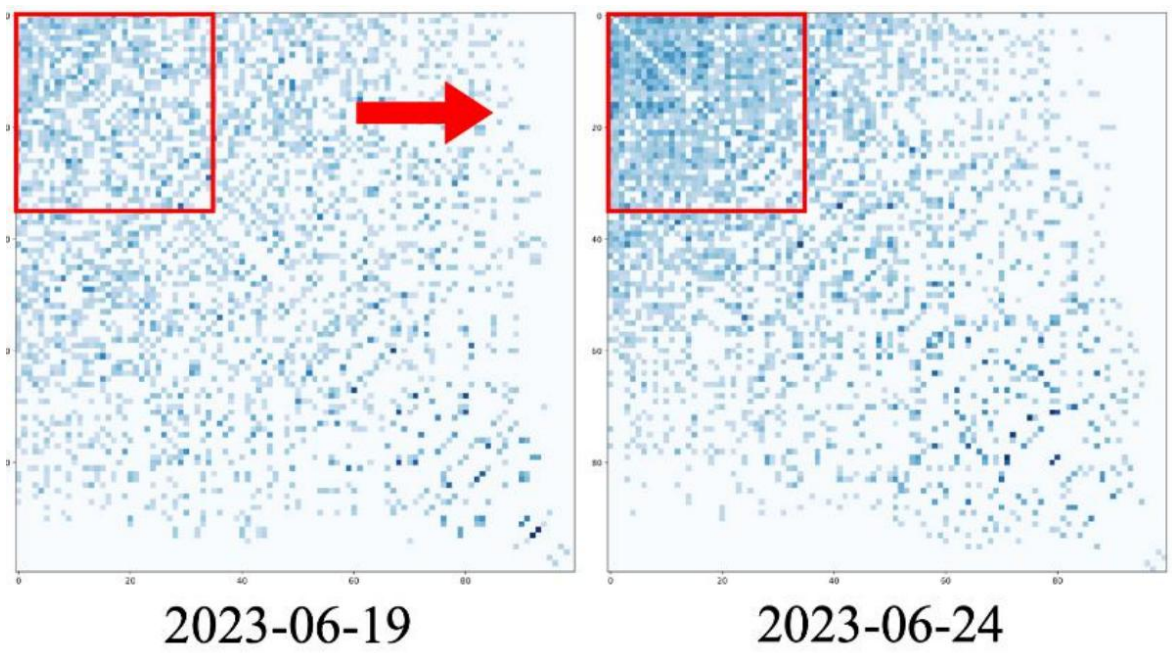
Experiment: inject a fake negative rumor into the social network

−27.5%

lower valuation by agents who saw the rumor

0.50 → 1.00

sell-to-buy ratio nearly doubles, and prices fall



Investors who think alike cluster tighter after the rumor: an echo chamber

No crash was programmed. Panic spread through reading, posting and copying.

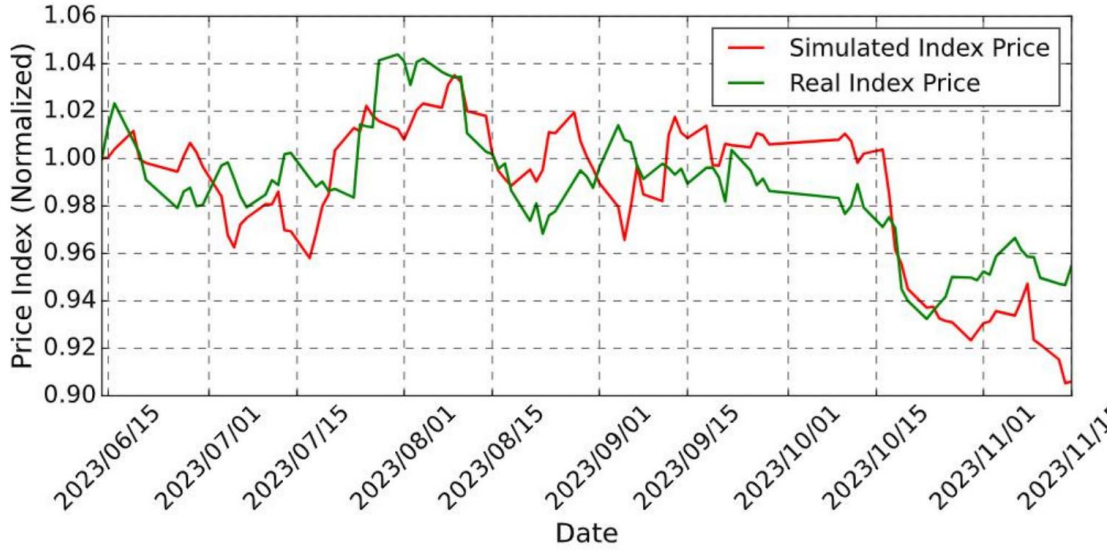
Figures & data: Yang et al., TwinMarket (NeurIPS 2025)

TwinMarket: a wind tunnel for financial markets

Test before it's real
Try rumors, policies and regulations on AI investors first

Crowds from individuals
Bubbles, panics and echo chambers emerge from 1,000 interacting agents

Limits
China A-shares only · one auction per day · AI agents ≠ real humans

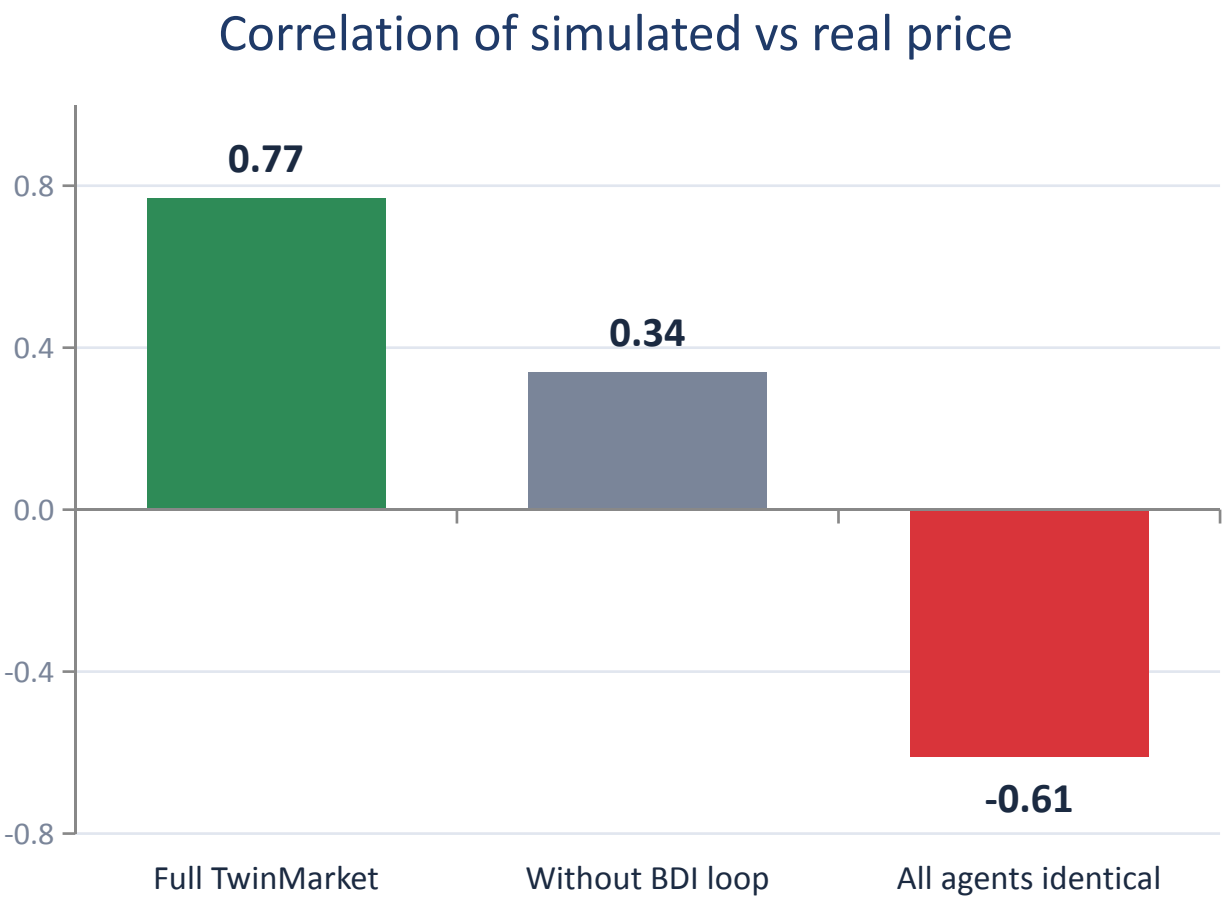


1,000-agent simulation vs real index

Thank you!

Figures & data: Yang et al., TwinMarket (NeurIPS 2025)

Backup: what if every investor were identical?



Variant	Corr. ↑	Kurtosis ↑	RMSE ↓
Full TwinMarket	0.77	5.24	0.02
Without BDI	0.34	4.25	0.07
All identical	−0.61	3.58	0.09

Diversity is the secret ingredient

Make every investor the same and the simulated market moves opposite to the real one.

Figures & data: Yang et al., TwinMarket (NeurIPS 2025)